Tracing Intertextuality at Scale
A Network Study of Poetic Influence

ABSTRACT (ENGLISH)

In this paper, I argue that intertextuality constitutes a particularly promising domain for distant reading research. The use of Text Reuse Detection (TRD) algorithms, in fact, makes it possible to map intertextual relations at scale and to identify regularities that can shed new light on the dynamics underlying literary influence. To test this approach on a broad and heterogeneous sample of literary production, I draw on the PoeTree collection and construct corpora of poetry in seven languages: Italian, English, Russian, French, Czech, German, and Hungarian. Applying TRD, I extract intertextual networks and compare selected structural features, with particular attention to the temporal reach of intertextual links and their distribution across authors. The results reveal significant corpus-specific differences in the temporal dimension of intertextuality. By contrast, the distribution of intertextual references among authors appears stable across corpora, converging on a heavy-tailed shape that is consistent with self-reinforcing, prestige-driven dynamics.

Keywords: Intertextuality, Distant Reading, Cultural Evolution, Prestige, Networks

ABSTRACT (ITALIANO)

Intertestualità su vasta scala e network di influenza fra poeti.

In questo contributo sostengo che l'intertestualità rappresenti un ambito di ricerca particolarmente promettente per il distant reading. L'impiego di algoritmi di Text Reuse Detection (TRD) consente infatti di mappare su larga scala le relazioni intertestuali e di individuarvi possibili regolarità, le quali potrebbero approfondire la nostra conoscenza delle dinamiche alla base dei processi di influenza letteraria. Al fine di ottenere un campione ampio e diversificato su cui testare un simile approccio, mi avvalgo della collezione PoeTree e costruisco corpora di poesie in sette lingue: italiano, inglese, russo, francese, ceco, tedesco e ungherese. L'applicazione di TRD mi permette di ricavare network intertestuali, di cui confronto alcune caratteristiche strutturali, in particolare l'estensione temporale dei legami intertestuali e la distribuzione degli stessi tra gli autori. I risultati evidenziano differenze significative tra i corpora per quanto riguarda la prima delle caratteristiche menzionate. La distribuzione di rimandi intertestuali tra autori sembra invece rimanere costante, convergendo su una forma 'heavy tailed' probabilmente causata dalla natura auto-rinforzantesi del prestigio letterario.

Parole chiave: Intertestualità, Distant Reading, Cultural Evolution, Prestigio, Networks

1. INTRODUCTION

As noted by Ciotti (2021), much research in the field of distant reading suffers from a mismatch between theory and methods, as the adoption of computational techniques often relies on literary-theoretical frameworks that, being grounded in interpretation and hermeneutics, lend themselves only partially to data-driven analysis. Intertextuality has been proposed (Roe, 2024: Rebora, in press) as a research area wherein such divide might be attenuated. From a methodological perspective, the detection of textual exchange (more or less explicit and exact) between literary works constitutes a straightforward computational problem, thus allowing for robust, reproducible analyses. From a theoretical perspective, the study of intertextual relations at scale may serve as the bedrock for developing a conception of literature as a complex evolving system. Such evolutionary perspective, already present in the seminal works by Moretti (2000a, 2003), has been recognized as a viable theoretical framework to ground distant reading reaserch (Ciotti, 2021), and is currently receiving increasing attention by scholars who combine computational approaches to the theories and models of cultural evolution (Konle et al., 2024; Sobchuk, 2023).

In this paper, I explore the use of Text Reuse Detection (TRD) as a computational framework to map intertextual relations in seven different corpora of Italian, English, Russian, French, Czech, German and Hungarian poems. The scope of my reaserch allows me to probe possible general dynamics of intertextual exchange, testing whether cross-corpora regularities can be found in: a) the average timespan between the publication of a poem and its subsequent intertextual reuses, and b) the distribution of instances of intertextual reuse across poets.

Intertextuality

The origin of the concept of intertextuality can be traced back to French post-structuralism, and particularly to the work of Julia Kristeva who, drawing from Saussurre and Bachtin, interpreted the literary text as "a permutation of texts, an intertextuality: in the space of a given text, several utterances, taken from other texts, intersect and neutralize one another" (Kristeva, 1980, p. 36) Famously aiming at downplaying the role of the author to foreground the text's own "productivity," post-structuralism proposed a radical view of intertextuality, conceptualizing it as the very condition for the emergence of literature. Radical, and, at the same time, somewhat vague: in Barthes, for instence, the intertextual field becomes a "general field of anonymous formulas, whose origin is rarely identifiable, of unconscious or automatic citations, given without quotation marks" (Barthes, 1990).

Such a loose conception of intertextuality hinders an operational take on the subject – indeed, how can one write an algorithm to identify formulas whose origin is, by definition, "rarely identifiable?" The vagueness of post-structuralist intertextuality has already been noted by several scholars, such as Genette (1982), who famously re-organized the concept under the label of "transtextuality." A more restrictive, yet also more easily operationalizable concept is what Juvan calls "citationality," namely a correspondence between literary works based on "certain 'objective' textual features" (Juvan, 2008, p. 142). Based on this, in my work I limit the scope of intertextuality to cases in which the relationship between hypotext and hypertext entails a clear and demonstrable exchange of textual material.

Text Reuse Detection

By conceptualizing intertextuality as "citationality" it becomes possible to operationaize it through TRD. TRD refers to a computational technique aimed at identifying "meaningful reiteration[s] of text, usually beyond the simple repetition of common language" (Romanello & Hengchen, 2021) in a corpus. Such reiterations needn't necessarily be verbatim quotes. In fact, TRD algorithms often capture even allusive or paraphrastic correspondences, similar to those "unconscious or automatic citations" mentioned by Barthes. Roe (2024) used this technique to uncover the influences behind the *Encyclopedie* by Diderot e D'Alembert, as Düring and colleagues (2023) did to map the intertextual relations within a corpus of historical journals.

As already evident in these examples, the automatic detection of text reuse is not usually an end in itself, but rather aims at mapping patterns of influence within a corpus. With regard to this, however, the focus of TRD on the sole literal level of the text constitutes a limitation, since not every case of literary influence results in the (more or less explicit) exchange of textual material. Nonetheless, it can be assumed that these 'invisible' cases occur at a steady rate across the corpus, so even though TRD gives a partial view of influence patterns, the degree of information lost should be consistent for every work. In other words, I am considering text reuse patterns as a *proxy* for influence dynamics.

Cultural Evolution and Literary Influence

Cultural evolution (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981) has been proposed as a framework to explain how cultural systems change over time. According to this perspective, change is the macro-level, emergent result of micro-level interactions between individuals, whereby cultural information is exchanged through biased social learning processes that tend to favour certain cultural traits over others. A similar evolutionary framework has been applied to the study of literature, by conceptualizing

literary forms as cultural traits transmitted through intertextual processes between authors (Moretti, 2000b; Sobchuk, 2023).

Tracing the network of intertextual relations in several distinct corpora may help uncover general patterns, possibly shedding light over the long-studied phenomenon of literary evolution (Tynianov, 2019). To this end, I opted for an explorative approach and decided to tackle the following reaserch questions:

1. Whether the average timespan between the publication of a poem and its subsequent intertextual reuses varies significantly across corpora. This research question is inspired by the work of Huges and colleagues (2012), who demonstrate that literary influence is strongly temporally localized, with most significant similarities occurring among authors active within a limited time window.
2. Whether the distribution of intertextual references across poets varies significantly across corpora. This research question is inspired by extensive research on citation networks (Barabàsi & Albert, 1999; Sheridan & Onodera, 2018) and literary fame (Martindale, 1995), which has shown that, across many domains, prestige dynamics tend to give rise to predictable distributions, whereby a small number of agents attract a disproportionate share of references.

2. MATERIALS AND METHODS

Data

My corpus is built on the basis of PoeTree (Plecháč et al., 2024), a multilingual collection of high-quality, metadata-rich poetic corpora accessible through a dedicated API. Given the rarity of cases of citationality, I anticipated corpus size to be a key parameter in my study. I therefore decided exclude the corpora with less than 10.000 poems, thus focusing only on the Hungarian, Russian, German, Italian, English, French, and Czech corpora.

Language	Version	# Poems	# Authors	# Tokens
Italian	original	39,749	243	11,073,882
	deduplicated	31,817	177	8,754,717
French	original	18,241	195	6,488,685
	deduplicated	18,067	194	6,421,757
Hungarian	original	12,953	48	2,560,232
	deduplicated	12,936	48	2,557,163
Russian	original	45,574	370	6,997,456
	deduplicated	45,488	368	6,987,947
English	original	41,806	447	15,039,289
	deduplicated	37,439	431	13,398,702
Czech	original	80,229	606	14,932,621
	deduplicated	70,798	606	13,241,243
German	original	74,351	258	13,458,630
	deduplicated	66,297	258	12,462,280

Table 1. Information on the corpora before and after deduplication.

From each of these, I first excluded all poems lacking a publication date. I then applied an additional filter for duplicates. Although PoeTree authors report that each corpus is deduplicated, this appears to apply only within the work of individual authors (that is, duplicates are identical poems *from the same author*). However, I identified numerous unflagged duplicates *between* authors. For example, the Italian corpus includes poems from both Malipiero's *Petrarca spirituale* and Leopardi's *Crestomazia italiana poetica*, which anthologize overlapping selections of Petrarch's work. Because such cases may confound automatic intertextuality detection, I personally deduplicated each corpus, treating as duplicates any poems with a Levenshtein similarity above 0.8 to an earlier poem, regardless of authorship.

Intertextual Networks

I performed TRD on each corpus individually using Passim (Romanello & Hengchen, 2021). Manual inspection of text reuse clusters[1] confirmed the versatility of the tool, capable of detecting not only verbatim quotes but also instances of parody or allusion.[2]

[1] In the context of TRD, a cluster is the set of documents that share the same passage.
[2] A telling example is the clearly parodical link identified by passim between Poe's *Annabel Lee* ("It was many and many a year ago, In a kingdom by the sea, That a maiden there lived whom you may know By the name of ANNABEL LEE; And this maiden she lived with no other thought Than to love and be loved by me. I was a child and she was a child, In this kingdom by the sea") and a poem by Robert F. Murray ("It was many and many a year ago, In a city by the sea, That a man there lived whom I happened to know By the name of Andrew M'Crie; And this man he slept in another room, But ground and had meals with me. I was an ass and he was an ass, In this city by the sea").

The inspection phase further revealed many clusters of intertextual relations between poems by the same pairs of authors. To prevent such tight reciprocal relationships from disproportionately shaping the overall network structure, I decided to retain only a single cluster per author group.

At this point I created, for each corpus, a directed network of intertextual relations by considering each poem as a node, with attributes encoding for its author and date of publication. I then used text reuse clusters to derive edges. For clusters with two elements, the latest poem was considered the source and the earliest the target of the edge. For clusters of size $n > 2$, n-1 edges were created, each connecting a 'later' poem (source) to the earliest (target).[3]

Poems in PoeTree corpora cover a vast period of time, ranging from the XIII to the XX century, with the distribution of poems per period varying significantly across corpora.[4] To allow for meaningful comparisons, I decided to limit my research to the period with the highest density of poems, namely 1750-1900, and sample 20.000 poems from each corpus, discarding those that didn't reach this number for the period in question. I then pruned the network of each of the remaining corpora by keeping only the edges whose source and target were included in the downsampled corpus. Networks that contained fewer than 10 edges after pruning were excluded, as such small samples would not support reliable statistical inference. As a result, only the English, German, and Russian corpora were retained for analysis.

Language	Clusters	Edges Before Downsampling	Edges After Downsampling
Italian	265	50	NA
French	7	4	NA
Hungarian	9	8	NA
Russian	41	32	22
English	60	42	21
Czech	54	40	9
German	319	120	18

Table 2. Text reuse clusters and number of edges in each corpus before and after downsampling.

Analysis

To test whether the average timespan between the publication of a poem and its subsequent intertextual reuses varies significantly across corpora, I computed, for each node in each network, the difference between the source's and target's publication dates. I then ran a Kruskal–Wallis test (since the data were non-normally distributed), with difference scores as the dependent variable (DV) and corpus as the independent variable (IV).

To address my second research question, I first aggregated each network's in-degree distribution by author; that is, I counted how many times each author of a poem in the network had been the target of an intertextual link. Authors in the corpus who never appear as the target of an intertextual link were assigned a value of 0.

I then tested whether the overall likelihood of being the target of an intertextual link varies across corpora. To do this, I created a binary variable encoding each author's quotation status (quoted vs. unquoted) and used a chi-square test of independence to compare quotation status across corpora.

Finally, I focused only on the intertextual networks (thus ignoring all non-quoted authors) and used a k-sample Anderson–Darling test to determine whether the nodes' in-degree distribution (aggregated by author) varies significantly across corpora.

3. RESULTS

The Kruskal–Wallis test revealed a significant effect of corpus on timespan ($H(2) = 10.5$, $p = .005$), indicating that the temporal scope of intertextual practices varies across corpora. The average timespan in

[3] In other words, for every passage shared by more than 2 works, I considered the earliest work as the original and all the others as 'imitators', discounting the possibility of mediated influence (work B quotes work A and work C quotes the same passage from work B). It should go without saying that this is a simplyfying, though necessary assumption. Note, however, that clusters of size > 2 turned out to be very rare, amounting to only 4% of the total.

[4] This is already made evident by the graph on the PoeTree web page: https://versologie.cz/poetree/general_info

the English (41 years, median = 39, SD = 25) and German (40 years, median = 35, SD = 33) corpora are considerably larger than the Russian corpus (18 years, median = 6, SD = 22). This result partially countradicts those of Hughes and colleagues (2012), who found that literary influence tends to be localized within relatively small time windows and decreases as these get wider. It must be noted, though, that their measure of literary influence was based on sylometric symilarity, which is *bound* to decrease monotonicaly with time, as a result of simple linguistic drift. By contrast, the prestige of certain poems or authors, and therefore the likelyhood that they will be the target of an intertextual link, can at times grow with time, according to a process that Moretti aptly described as a "low of increasing return" (Moretti 2000b).

Proof of the presence of general dynamics regulating intertextual patterns emerged from the investigation of my second research question, which found no significant variation in the distribution of intertextual links across corpora. The chi-square test ($\chi^2(2) = 1.3$, $p = .5$) indicates that the likelihood of an author being quoted is very small (5% in the English and Russian corpora, 8% in the German corpus), and does not vary significantly across corpora. The k-sample Anderson–Darling test also did not reach conventional significance ($A^2 = -0.8$, $p = .6$) suggesting that the distributions of authors' in-degree counts are not significantly different across corpora. It is important to consider that the relatively small sample sizes[5] may have limited the power of this test. Nonetheless, the distributions clearly display a common heavy-tailed behavior, with most authors receiving only one or two intertextual links and only a few receiving more, with a maximum of 4 in the English (Wordsworth) and German (Schiller and Herder) corpora, and of 5 in the Russian corpus (Zhukovskij).

4. DISCUSSION

Intertexruality is a promising research area to bridge literary theory and computational approaches. By conceptualizing it as citationality (Juvan, 2008), in fact, it becomes possible to operationalize it as a TRD problem and study it at scale. This approach has been already proposed in recent research (Roe, 2024; Rebora, in press), to which my work adds further proof of its feasibility. However, my study also raises possible concerns. As data show, explicit text reuse between literary works is very rare. Conceptually, this doesn't necessarily pose an issue, as it can be assumed that, though text reuse networks are only a partial sample taken from larger intertextual networks, they still constitute a valid proxy for assessing their structural properties. Pragmatically, though, the rarity of text reuse limits the applicability of this method, as it requires very large corpora, from which a sufficient number of text reuse links can be extracted to allow for statistical inference.

The investigation of cross-corpora dynamics gave mixed results. Corpus-specific patterns were observed with regard to average timespan between intertextually linked poems. This finding indicates the absence of rigid time constraints on intertextuality, that is, the relative influence that old and contemporary authors may exert on a poem is contingent and context-specific.

By contrast, the distribution of intertextual links amongst authors follows a constant pattern: on the one hand, the likelyhood of being the target of an intertextual link is very small, but constant across corpora; on the other hand, all distributions of intertextual links across authors converge onto a 'heavy tail' shape, with many rarely quoted authors and a low number of highly quoted ones. A similar behavior has been observed in other areas, such as citantions in scientific papers (Barabàsi & Albert, 1999), sales figures across books (Fenner et al., 2010), and crtical mentions of poets (Martindale, 1995). What these phenomena have in common is their reliance on the self-feeding loop of prestige dynamics, whereby highly successful works or authors are more likely than others to acqiure additional success. More than historically-determined factors, this dynamic seems to rely on the gregarious nature of human attention-attribution strategies (we tend to pay more attention to what others are paying attention to) and practical matters of visibility (highly successful authors have more chances to be read, and therefore quoted). My data can therefore be interpreted as suggesting the presence of a universal prestige-based dynamic (Henrich & Gil-White, 2001; Jiménez & Mesoudi, 2019) regulating intertextual exchange and influence dynamics.

[5] 15 authors in the English corpus, 17 in the Russian one and 12 in the German one.

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